

SUKHADA GHEWARI

Applied ML Researcher and Engineer focused on AI for Climate and Health

 [Website](#)  [Email](#)  [LinkedIn](#)  [GitHub](#)

Education

University of California San Diego

Master of Science in Computer Science | GPA: 3.97/4.0

Sep 2021 – Mar 2023

San Diego, USA

Walchand College of Engineering

Bachelor of Technology in Computer Science | CGPA: 9.51/10 (Gold Medalist)

Jul 2015 – May 2019

Sangli, India

Experience

Independent Researcher

Jul 2026 – Present | Pune, India

- **Cyclone rapid intensification**: built an machine learning pipeline on **around 80,000 observations from 4,000+ storms** (IBTrACS, ERA5, 1980–2025) to predict 30+ kt jumps in peak winds within 24 h, using 5-fold cross-validated feature ablations
- Curated **around 80,000 observations from 4,000+ storms** (IBTrACS, 1980–2025) with **ERA5** environmental and satellite features; selected features and models via **5-fold cross-validated ablations**, showing that environmental predictors add meaningful skill
- On held-out 2016–2025 storms, **XGBoost** achieves **0.42 AUPRC** (about 4× the no-skill baseline) and **0.86 ROC-AUC**, detecting **about three quarters** of rapid-intensification events at an F2-optimised threshold
- **Forest Sentinel** (AmazôniaHack): offline prototype that drafts traceable environmental inspection reports from field voice notes, photos and notes in low-resource setting (Whisper, OCR, Qwen3-VL), running on a laptop CPU with no paid APIs

Software Engineer 2 | Oracle

May 2023 – Dec 2025 | Redwood City, USA

- Led development of an LLM-powered SQL conversion tool that migrated **200K+ queries** (Vertica/Snowflake to Oracle SQL) across data warehouses of **500+ enterprise hospital clients**, combining Meta Llama 3.1 with RAG, prompt engineering, and RegEx heuristics; achieved an **85%+ conversion success rate**, saving hundreds of hours of manual consultant effort per migration cycle
- Automated end-to-end client onboarding for HealtheAnalytics, a clinical analytics product, by orchestrating multi-step workflows across microservices for 500+ clients, eliminating manual provisioning effort

Software Development Intern | Oracle

Jun 2022 – Sep 2022 | Remote

- Built and productionized an **auto-healing recommendation system** for CI/CD pipeline failures, using NLP (TF-IDF, stemming, lemmatization) and Gradient Boosting to recommend fixes, reducing time-to-resolution for recurring errors

Graduate Student Researcher | Ideker Lab, UC San Diego

Oct 2021 – Jun 2022 | La Jolla, CA

- Built a Synapse Protein Ontology from protein interaction networks using Random Forest and HiDef community detection, forming a hierarchical ontology used as an interpretable neural network for psychiatric disease prediction (FDR ≤ 0.05)

Researcher | TCS Research (TRDDC)

Jul 2019 – May 2021 | Pune, India

- Developed a Python tool to detect bias in state-of-the-art face recognition (FaceNet) and face attribute detection models on CelebA, measuring performance gaps across gender and age subgroups with fairness metrics (FPR, FNR, accuracy parity)
- Authored a research paper on a Domain Name Generation Algorithm (CFG + unsupervised learning) for defensive registration against phishing/DNS abuse -published at eCrime 2021

Projects

Crop Type Mapping in Kenya | Python, ML, Sequence Models

Spring 2022

- Built a multi-class crop type classifier on Sentinel-2 satellite time-series data (14 spectral bands, 5,700+ fields) for smallholder farms in Kenya, benchmarking 5 ML models and achieving 65.3% accuracy with Gradient Boosting

Skills

Languages: Python, Java, SQL

Machine Learning: Deep Learning, NLP, Computer Vision, LLMs, RAG

Frameworks & Tools: PyTorch, Scikit-learn, PySpark, CI/CD, Docker, Git, AWS, OCI

Research

- **Ghewari S.**, Deshpande S. *Which Information Matters for Rapid Intensification of Tropical Cyclones? A Global-Basin Feature Ablation Study* **NeurIPS 2026** – Women in Machine Learning (WiML) Workshop, Sydney, Australia (Accepted)
- Kumar N., **Ghewari S.**, et al. *When Diversity Meets Hostility: A Study of Domain Squatting Abuse in Online Banking.* **eCrime 2021** – IEEE Symposium on Electronic Crime Research (Filed related patent in India)